

ENGR 3323: Signals and Systems

HW 9_Ch6 Answer Keys

Q1)

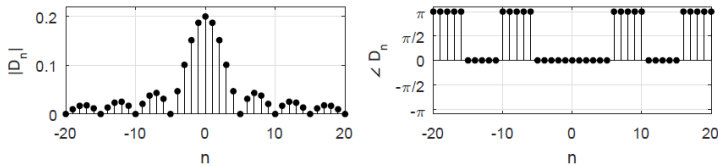
a) for $x(t)$

$$D_n = \frac{1}{5} \text{sinc}\left(\frac{n\pi}{5}\right)$$

for $y(t)$

$$D_n = \frac{1}{2\pi} \int_0^{2\pi} \frac{t}{2\pi} e^{-jnt} dt = \frac{j}{2\pi n},$$

b)

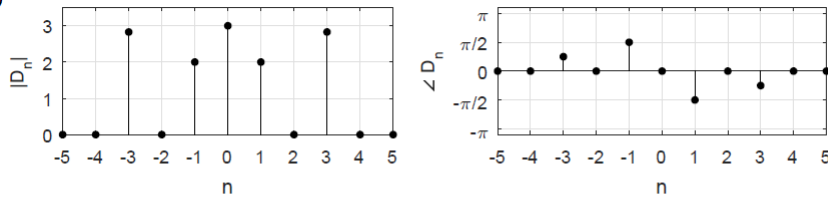


c) For $x(t)$, the 5th harmonic will be less than 10% of the DC and the approximate power for the DC and the first five harmonics is $P = 0.1805$.

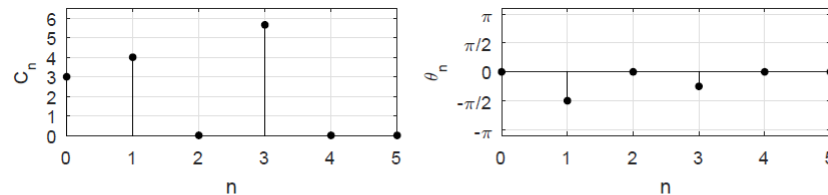
For $y(t)$, the 2nd harmonic will be less than 10% of the DC and the approximate power for the DC and the first two harmonics is $P = 0.3013$.

Q2)

a)



b) $x(t) = 3 + 4 \cos\left(t - \frac{\pi}{2}\right) + 4\sqrt{2} \cos\left(3t - \frac{\pi}{4}\right)$

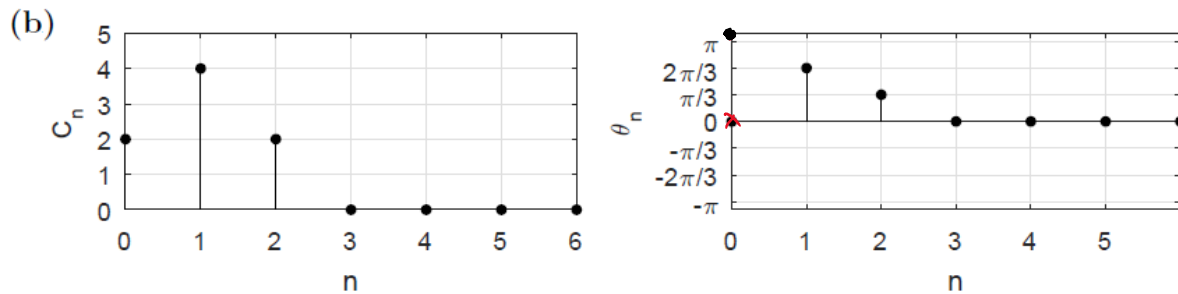


c) Since, the trigonometric series in part (b) is obtained from the exponential series in part (a), the two series are equivalent.

d) The lowest frequency in the spectrum is 0 and the highest frequency is 3 rad/s. Therefore, the bandwidth is 3 rad/s or $\frac{3}{2\pi}$ Hz.

Q3)

$$(a) \quad x(t) = -2 + 2e^{j(t+\frac{2\pi}{3})} + 2e^{-j(t+\frac{2\pi}{3})} + e^{j(2t+\frac{\pi}{3})} + e^{-j(2t+\frac{\pi}{3})}$$



$$(c) \quad x(t) = -2 + 4 \cos\left(t + \frac{2\pi}{3}\right) + 2 \cos\left(2t + \frac{\pi}{3}\right)$$

$$(d) \quad x(t) = -2 + 4 \cos\left(t + \frac{2\pi}{3}\right) + 2 \cos\left(2t + \frac{\pi}{3}\right) \\ = -2 + 2e^{j(t+\frac{2\pi}{3})} + 2e^{-j(t+\frac{2\pi}{3})} + e^{j(2t+\frac{\pi}{3})} + e^{-j(2t+\frac{\pi}{3})}$$

Q4 For $x(t) = (2 + j2)e^{j3t} + j2e^{jt} + 3 - j2e^{jt} + (2 - j2)e^{j3t}$

$$D_{-3} = 2 + j2; \quad D_{-2} = 0; \quad D_{-1} = j; \quad D_0 = 3; \quad D_1 = -j; \quad D_2 = 0; \quad D_3 = 2 - j2$$

$$H(j\omega) = \frac{j\omega}{(-\omega^2 + 3) + j2\omega}$$

$$y(t) = \sum_{n=-\infty}^{\infty} D_n H(jn\omega_0) e^{jn\omega_0 t} = \sqrt{2} \cos(t - \pi/4) + \cos(3t - \pi/2)$$

Q5)

$$(a) \quad x(t) = \sum_{n=-\infty}^{\infty} \frac{1 - e^{-1}}{1 + j2\pi n} e^{j2\pi n t}$$

$$(b) \quad H(j\omega) = \frac{j\omega}{j\omega + 1}$$

$$y(t) = \sum_{n=-\infty}^{\infty} D_n H(j2\pi n) e^{j2\pi n t} \\ = \sum_{n=-\infty}^{\infty} \frac{j2\pi n(1 - e^{-1})}{(1 + j2\pi n)^2} e^{j2\pi n t}$$